1 I&I for a sinusoidal system

The **I&I method** (**Immersion and Invariance**) is an adaptive control method that estimates the dynamics of unmeasured states. The method first specifies the desired dynamics of the error z, then derives adaptation laws that produce those dynamics.

1.1 Problem formulation

Consider the measurable signal

$$y(t) \;=\; A\sin(\omega t+\phi)$$

with unknown A, ω and ϕ.

We obtain

$$\begin{aligned} y' &= \omega A\cos(\omega t+\phi) \\ y'' &= -\omega^2 A\sin(\omega t+\phi) = -\omega^2 y \end{aligned}$$

Let $x = \omega A\cos(\omega t+\phi)$ and $\theta = \omega^2$. To determine the angular frequency ω, consider the following linear system

$$\begin{aligned} \dot{y} &= x \\ \dot{x} &= -\theta\, y \\ \dot{\theta} &= 0 \end{aligned}$$

where y is measurable and θ and x are unknown.

Once θ has been estimated, if $\omega > 0$, set

$$\omega \;=\; \sqrt{\theta}$$

For the two parameters A and ϕ, observe that

$$\begin{aligned} y^2+\Big(\frac{x}{\omega}\Big)^2 &= A^2\sin^2(\omega t+\phi)+\left(\frac{\cancel{\omega} A\cos(\omega t+\phi)}{\cancel{\omega}}\right)^2 \\ &= A^2\sin^2(\omega t+\phi)+A^2\cos^2(\omega t+\phi) \\ &= A^2(\sin^2(\omega t+\phi)+\cos^2(\omega t+\phi)) \\ &= A^2 \end{aligned}$$

and that

$$\begin{aligned} \omega\frac{y}{x} &= \cancel{\omega}\frac{\cancel{A}\sin(\omega t+\phi)}{\cancel{\omega}\cancel{A}\cos(\omega t+\phi)} \\ &= \frac{\sin(\omega t+\phi)}{\cos(\omega t+\phi)} \\ &= \tan(\omega t+\phi) \end{aligned}$$

It follows that

$$\begin{aligned} A &= \sqrt{y^2+\Big(\frac{x}{\omega}\Big)^2} \\ \phi &= \tan^{-1}\Big(\omega\frac{y}{x}\Big) \end{aligned}$$

1.2 Estimation of the angular frequency ω

We now estimate the angular frequency ω. As shown in the previous section, determining the angular frequency w requires estimating x and $\theta = \omega^2$ in the system

$$\begin{aligned} \dot{y} &= x \\ \dot{x} &= -\theta\, y \\ \dot{\theta} &= 0 \end{aligned}$$

To do so, the I&I method introduces the auxiliary dynamic variable $\xi = \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix}$ and defines the estimation errors for x and θ as z_x and z_θ, respectively. Since the dynamics of x and θ are coupled, set

$$\begin{aligned} z_x &= \beta_1(y, \xi_1) + h_1(\xi_1, \theta) - x \\ z_\theta &= \beta_2(y, \xi_1, \xi_2) - \theta \end{aligned}$$

where $\beta_1, \beta_2, h_1 \in C^1$. For simplicity, the auxiliary variable ξ_2 appears only in the second equation, which omits the function h_2.

The dynamics of the estimation error for x are therefore given by

$$\begin{aligned} \dot{z}_x &= \frac{d\beta_1(y,\xi_1)}{d\,y}\dot{y} + \frac{d\beta_1(y,\xi_1)}{d\xi_1}\dot{\xi}_1 + \frac{dh_1(\xi_1,\theta)}{d\,\xi_1}\dot{\xi}_1 + \frac{dh_1(\xi_1,\theta)}{d\theta}\dot{\theta} - \dot{x} \\ &= \frac{d\beta_1(y,\xi_1)}{d\,y}x + \frac{d\beta_1(y,\xi_1)}{d\xi_1}\dot{\xi}_1 + \frac{dh_1(\xi_1,\theta)}{d\,\xi_1}\dot{\xi}_1 + \theta\, y \end{aligned}$$

Recall that

$$\begin{aligned} z_x &= \beta_1(y, \xi_1) + h_1(\xi_1, \theta) - x \\ &\Downarrow \\ x &= \beta_1(y, \xi_1) + h_1(\xi_1, \theta) - z_x \end{aligned}$$

so

$$\begin{aligned} \dot{z}_x &= \frac{d\beta_1(y,\xi_1)}{d\,y}x + \frac{d\beta_1(y,\xi_1)}{d\xi_1}\dot{\xi}_1 + \frac{dh_1(\xi_1,\theta)}{d\,\xi_1}\dot{\xi}_1 + \theta\, y \\ &= \frac{d\beta_1(y,\xi_1)}{d\,y}(\beta_1(y,\xi_1) + h_1(\xi_1,\theta) - z_x) + \frac{d\beta_1(y,\xi_1)}{d\xi_1}\dot{\xi}_1 + \frac{dh_1(\xi_1,\theta)}{d\,\xi_1}\dot{\xi}_1 + \theta\, y \\ &= \frac{d\beta_1(y,\xi_1)}{d\,y}(\beta_1(y,\xi_1) + h_1(\xi_1,\theta)) - \frac{d\beta_1(y,\xi_1)}{d\,y}z_x + \frac{d\beta_1(y,\xi_1)}{d\xi_1}\dot{\xi}_1 + \frac{dh_1(\xi_1,\theta)}{d\,\xi_1}\dot{\xi}_1 + \theta\, y \\ &= -\frac{d\beta_1(y,\xi_1)}{d\,y}z_x + \frac{d\beta_1(y,\xi_1)}{d\,y}(\beta_1(y,\xi_1) + h_1(\xi_1,\theta)) + \frac{d\beta_1(y,\xi_1)}{d\xi_1}\dot{\xi}_1 + \frac{dh_1(\xi_1,\theta)}{d\,\xi_1}\dot{\xi}_1 + \theta\, y \end{aligned}$$

To make the dynamics of the estimation error for x, given by $\dot{z_x}$, depend only on z_x, impose

$$\frac{d\beta_1(y,\xi_1)}{d\,y}(\beta_1(y,\xi_1) + h_1(\xi_1,\theta)) + \frac{d\beta_1(y,\xi_1)}{d\xi_1}\dot{\xi}_1 + \frac{dh_1(\xi_1,\theta)}{d\,\xi_1}\dot{\xi}_1 + \theta\, y = 0$$

In that case,

$$\dot{z}_x = -\frac{d\beta_1(y,\xi_1)}{d\,y}z_x$$

To make the estimation error for x exponentially stable, set

$$\begin{aligned}\frac{d\beta_1(y,\xi_1)}{d\,y} &= k\\ &\Downarrow\\ \beta_1(y,\xi_1) &= k\,y+f(\xi_1)\end{aligned}$$

It follows that

$$\begin{aligned}\frac{d\beta_1(y,\xi_1)}{d\,y}(\beta_1(y,\xi_1)+h_1(\xi_1,\theta))+\frac{d\beta_1(y,\xi_1)}{d\xi_1}\dot{\xi}_1+\frac{dh_1(\xi_1,\theta)}{d\,\xi_1}\dot{\xi}_1+\theta\,y &= 0\\ k\,(k\,y+f(\xi_1))+k\cdot h_1(\xi_1,\theta)+\frac{d\,f(\xi_1)}{d\xi_1}\dot{\xi}_1+\frac{dh_1(\xi_1,\theta)}{d\,\xi_1}\dot{\xi}_1+\theta\,y &= 0\end{aligned}$$

For simplicity, set $f(\xi_1)=0$ to obtain

$$\begin{aligned}k\left(k\,y+\cancel{f(\xi_1)}\right)+k\cdot h_1(\xi_1,\theta)+\frac{d\,f(\xi_1)}{\cancel{d\xi_1}}\dot{\xi}_1+\frac{dh_1(\xi_1,\theta)}{d\,\xi_1}\dot{\xi}_1+\theta\,y &= 0\\ k^2y+k\cdot h_1(\xi_1,\theta)+\frac{dh_1(\xi_1,\theta)}{d\,\xi_1}\dot{\xi}_1+\theta\,y &= 0\end{aligned}$$

Note that $\frac{dh_1(\xi_1,\theta)}{d\,\xi_1}\neq 0$, since otherwise

$$\begin{aligned}k^2y+k\cdot h_1(\xi_1,\theta)+\theta\,y &= 0\\ k\cdot h_1(\xi_1,\theta) &= -(k^2+\theta)y\\ h_1(\xi_1,\theta) &= -\frac{(k^2+\theta)}{k}y\end{aligned}$$

and hence $h_1(\xi_1,\theta)$ would depend on y. To simplify the calculations, set

$$h_1(\xi_1,\theta) = A\,\xi_1$$

with constant A. This gives

$$\begin{aligned}k^2y+k\cdot h_1(\xi_1,\theta)+\frac{dh_1(\xi_1,\theta)}{d\,\xi_1}\dot{\xi}_1+\theta\,y &= 0\\ (k^2+\theta)y+k\cdot A\,\xi_1+A\dot{\xi}_1 &= 0\end{aligned}$$

Setting $A=(k^2+\theta)$ allows us to factor out A, giving

$$\begin{aligned}(k^2+\theta)y+k(k^2+\theta)\xi_1+(k^2+\theta)\dot{\xi}_1 &= 0\\ (k^2+\theta)(y+k\xi_1+\dot{\xi}_1) &= 0\end{aligned}$$

To satisfy this equation,

$$\begin{aligned}y+k\xi_1+\dot{\xi}_1 &= 0\\ \dot{\xi}_1 &= -k\xi_1-y\end{aligned}$$

In summary,

$$\begin{aligned}z_x &= \beta_1(y,\xi_1)+h_1(\xi_1,\theta)-x\\ &= k\,y+(k^2+\theta)\xi_1-x\\ \dot{\xi}_1 &= -k\xi_1-y\end{aligned}$$

Note that, since

$$\xi_1(t) \;=\; e^{-\mathrm{kt}}\xi_1(0) + \int_0^t e^{-k(t-\tau)}(-y)\,d\tau$$

and $-A \le y \le A \quad \Rightarrow \quad -A \le -y \le A$,

$$\begin{aligned}
\xi_1(t) &= e^{-\mathrm{kt}}\xi_1(0) + \int_0^t e^{-k(t-\tau)}(-y)\,d\tau \\
&\le e^{-\mathrm{kt}}\xi_1(0) + \int_0^t e^{-k(t-\tau)}A\,d\tau \\
&= e^{-\mathrm{kt}}\xi_1(0) + A\frac{e^{-k(t-\tau)}}{k}\Big|_0^t \\
&= e^{-\mathrm{kt}}\xi_1(0) + A\frac{1-e^{-kt}}{k}\Big|_0^t \\
&< \infty
\end{aligned}$$

We now derive the dynamics of the estimation error for θ.

$$z_\theta \;=\; \beta_2(y,\xi_1,\xi_2) - \theta$$

We obtain

$$\begin{aligned}
\dot{z}_\theta &= \frac{d\beta_2(y,\xi_1,\xi_2)}{dy}\dot{y} + \frac{d\beta_2(y,\xi_1,\xi_2)}{d\xi_1}\dot{\xi}_1 + \frac{d\beta_2(y,\xi_1,\xi_2)}{d\xi_2}\dot{\xi}_2 - \cancel{\dot{\theta}} \\
&= \frac{d\beta_2(y,\xi_1,\xi_2)}{dy}x + \frac{d\beta_2(y,\xi_1,\xi_2)}{d\xi_1}(-k\xi_1 - y) + \frac{d\beta_2(y,\xi_1,\xi_2)}{d\xi_2}\dot{\xi}_2
\end{aligned}$$

where

$$\begin{aligned}
z_x &= k\,y + (k^2+\theta)\xi_1 - x \\
z_\theta &= \beta_2(y,\xi_1,\xi_2) - \theta \\
&\Downarrow \\
x &= k\,y + (k^2+\theta)\xi_1 - z_x \\
\theta &= \beta_2(y,\xi_1,\xi_2) - z_\theta \\
&\Downarrow \\
x &= k\,y + (k^2 + \beta_2(y,\xi_1,\xi_2) - z_\theta)\xi_1 - z_x \\
&= k\,y + k^2\xi_1 + \beta_2(y,\xi_1,\xi_2)\xi_1 - z_\theta\xi_1 - z_x
\end{aligned}$$

Substituting gives

$$\begin{aligned}
\dot{z}_\theta &= \frac{d\beta_2(y,\xi_1,\xi_2)}{dy}x + \frac{d\beta_2(y,\xi_1,\xi_2)}{d\xi_1}(-k\xi_1 - y) + \frac{d\beta_2(y,\xi_1,\xi_2)}{d\xi_2}\dot{\xi}_2 \\
&= \frac{d\beta_2(y,\xi_1,\xi_2)}{dy}(k\;y + k^2\xi_1 + \beta_2(y,\,\xi_1,\,\xi_2)\xi_1 - z_\theta\xi_1 - z_x) + \frac{d\beta_2(y,\xi_1,\xi_2)}{d\xi_1}(-k\xi_1 - y) + \\
&\quad \frac{d\beta_2(y,\xi_1,\xi_2)}{d\xi_2}\dot{\xi}_2 \\
&= -\frac{d\beta_2(y,\xi_1,\xi_2)}{dy}z_\theta\xi_1 - \frac{d\beta_2(y,\xi_1,\xi_2)}{dy}z_x + \frac{d\beta_2(y,\xi_1,\xi_2)}{dy}(k\;y + k^2\xi_1 + \beta_2(y,\,\xi_1,\,\xi_2)\xi_1) + \\
&\quad \frac{d\beta_2(y,\xi_1,\xi_2)}{d\xi_1}(-k\xi_1 - y) + \frac{d\beta_2(y,\xi_1,\xi_2)}{d\xi_2}\dot{\xi}_2
\end{aligned}$$

To make this equation depend only on z_x and z_θ, impose

$$\frac{d\beta_2(y,\xi_1,\xi_2)}{dy}(k\,y+k^2\xi_1+\beta_2(y,\xi_1,\xi_2)\xi_1)+\frac{d\beta_2(y,\xi_1,\xi_2)}{d\xi_1}(-k\xi_1-y)+\frac{d\beta_2(y,\xi_1,\xi_2)}{d\xi_2}\dot{\xi}_2 \;=\; 0$$

In that case,

$$\dot{z}_\theta \;=\; -\frac{d\beta_2(y,\xi_1,\xi_2)}{dy}z_x-\frac{d\beta_2(y,\xi_1,\xi_2)}{dy}\xi_1 z_\theta$$

To obtain $z_\theta \to 0$, set

$$\begin{aligned}\frac{d\beta_2(y,\xi_1,\xi_2)}{dy} &= \xi_1\\ &\Downarrow\\ \beta_2(y,\xi_1,\xi_2) &= \xi_1 y+f(\xi_2)\end{aligned}$$

Then

$$\begin{aligned}\dot{z}_\theta &= -\xi_1 z_x-\xi_1^2 z_\theta\\ &\to -\xi_1^2 z_\theta\end{aligned}$$

because $\xi_1 \in L_\infty$ and $z_x \to 0$. Therefore, z_θ converges exponentially to zero.

Finally, we determine the conditions under which the equation

$$\frac{d\beta_2(y,\xi_1,\xi_2)}{dy}(k\,y+k^2\xi_1+\beta_2(y,\xi_1,\xi_2)\xi_1)+\frac{d\beta_2(y,\xi_1,\xi_2)}{d\xi_1}(-k\xi_1-y)+\frac{d\beta_2(y,\xi_1,\xi_2)}{d\xi_2}\dot{\xi}_2 \;=\; 0$$

holds. We obtain

$$\begin{aligned}\frac{d\beta_2(y,\xi_1,\xi_2)}{dy}(k\,y+k^2\xi_1+\beta_2(y,\xi_1,\xi_2)\xi_1)+\frac{d\beta_2(y,\xi_1,\xi_2)}{d\xi_1}(-k\xi_1-y)+\frac{d\beta_2(y,\xi_1,\xi_2)}{d\xi_2}\dot{\xi}_2 &= 0\\ \xi_1[k\,y+k^2\xi_1+(\xi_1 y+f(\xi_2))\xi_1]+y(-k\xi_1-y)+\frac{d\,f_2(\xi_2)}{d\xi_2}\dot{\xi}_2 &= 0\\ \xi_1 k\,y+k^2\xi_1^2+(\xi_1 y+f(\xi_2))\xi_1^2+y(-k\xi_1-y)+\frac{d\,f_2(\xi_2)}{d\xi_2}\dot{\xi}_2 &= 0\end{aligned}$$

Setting $f_2(\xi_2)=\xi_2$ gives

$$\begin{aligned}\xi_1 k\,y+k^2\xi_1^2+(\xi_1 y+\xi_2)\xi_1^2+y(-k\xi_1-y)+\dot{\xi}_2 &= 0\\ \xi_1\cancel{k}\,y+k^2\xi_1^2+\xi_1^3 y+\xi_1^2\xi_2-\cancel{k}\xi_1 y-y^2+\dot{\xi}_2 &= 0\\ \dot{\xi}_2 &= -k^2\xi_1^2-\xi_1^3 y-\xi_1^2\xi_2+y^2\\ \dot{\xi}_2 &= -\xi_1^2(k^2+\xi_1 y+\xi_2)+y^2\end{aligned}$$

In summary,

Theorem 1. *Consider the system*

$$\begin{aligned}\dot{y} &= x\\ \dot{x} &= -\theta y\\ \dot{\theta} &= 0\end{aligned}$$

where

$$y(t) = A\sin(\omega t + \phi) \qquad \theta = \omega^2$$

If

$$\begin{aligned} \hat{\theta} &= \xi_1 y + \xi_2 \\ \hat{x} &= k\,y + (k^2 + \hat{\theta})\xi_1 \end{aligned}$$

where

- $k > 0$ *is a free design parameter*
- ξ_1 *and* ξ_2 *are auxiliary variables with dynamics given by*

$$\begin{aligned} \dot{\xi}_1 &= -k\,\xi_1 - y \\ \dot{\xi}_2 &= -\xi_1^2(k^2 + \xi_1 y + \xi_2) + y^2 \end{aligned}$$

then the estimates converge asymptotically to the true parameter values.

$$\begin{aligned} \lim_{t\to\infty} \hat{\theta}(t) &= \theta \\ \lim_{t\to\infty} \hat{x}(t) &= x \end{aligned}$$

2 Assignment 4

Given the two sinusoidal signals

$$y_1(t) = \sin(100t), \qquad y_2(t) = \sin t$$

implement the I&I frequency estimator and run numerical simulations to show how well it estimates the angular frequency of each signal.

Then analyse the effect of the design parameter k. Run simulations for different gain values and discuss a suitable choice in terms of convergence speed, estimation accuracy, and transient behaviour.

Finally, consider the effect of a periodic disturbance on the signal measurement. Assume the measured signal is

$$y_1(t) + d(t),$$

where

$$d(t) = 0.001\sin(50t).$$

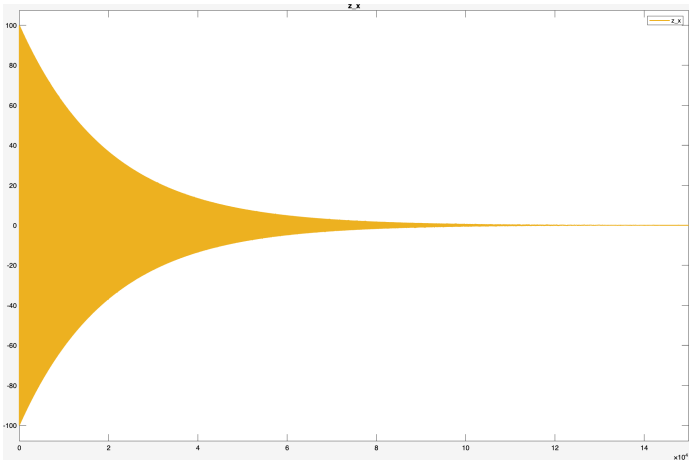
Run simulations for this case as well. Discuss how the disturbance affects convergence and frequency estimation accuracy by comparing the results with the disturbance-free case.

3 $y_1 = \sin 100t$

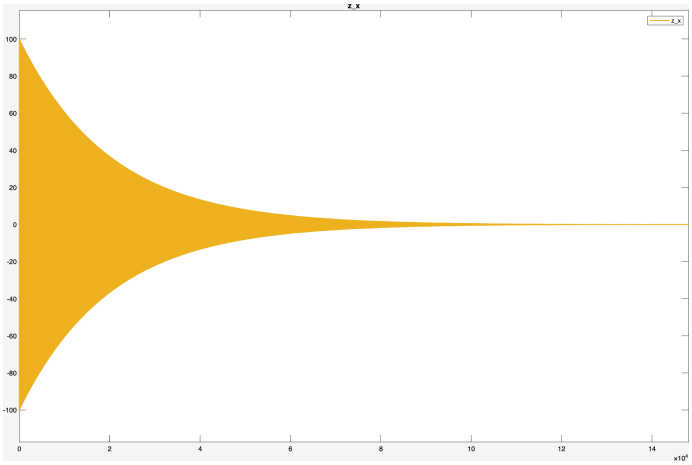
3.1 Estimation error z_x

With a disturbance-free measurement of $y_1 = \sin(100t)$, the estimate of the angular frequency ω behaves as follows.

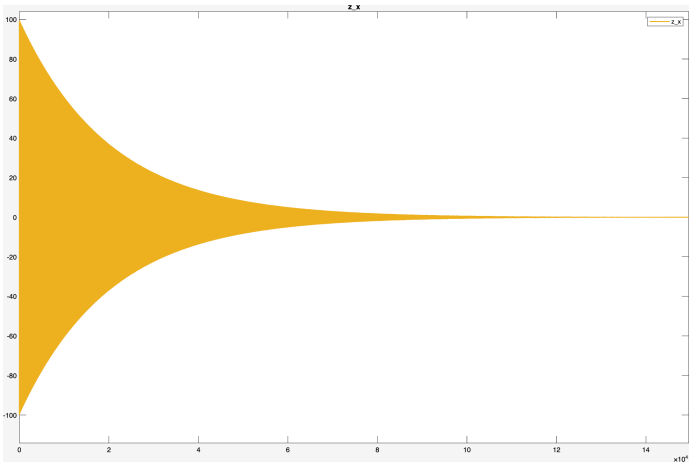
- For $k = 0.1$, it converges in about $1.4 \cdot 10^5 s$ after numerous oscillations with a maximum amplitude of 100.



- For $k = 1$, it converges in about $1.4 \cdot 10^5 s$, with oscillations reaching an amplitude of 100.



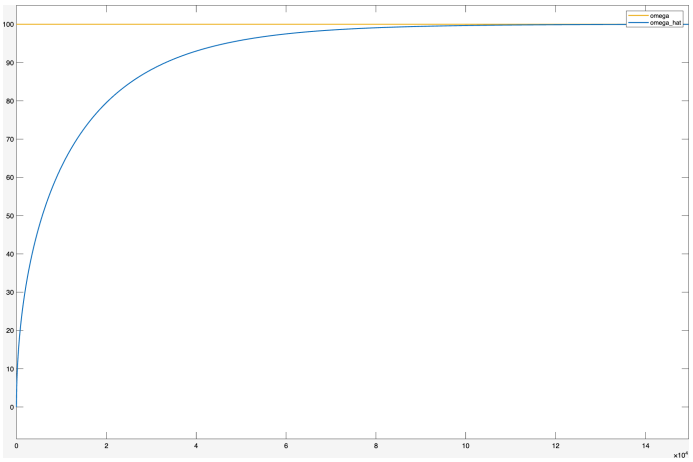
- For $k = 10$, it converges in about $1.3 \cdot 10^5 s$, with oscillations reaching an amplitude of 100.



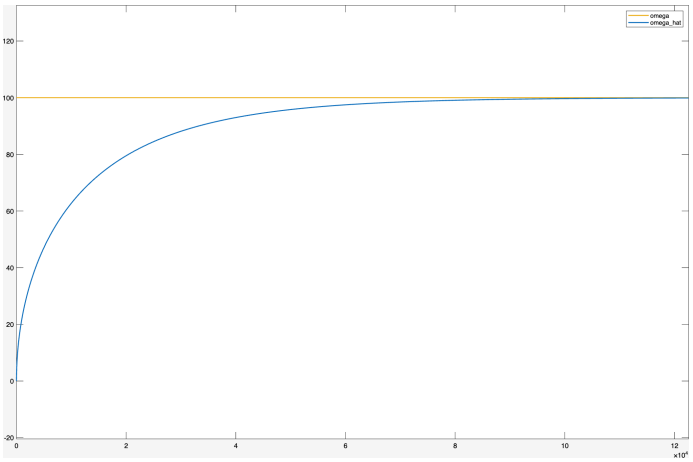
3.2 Estimation of the angular frequency ω

With a disturbance-free measurement of $y_1 = \sin(100t)$, the estimation error for x, denoted by z_x, behaves as follows.

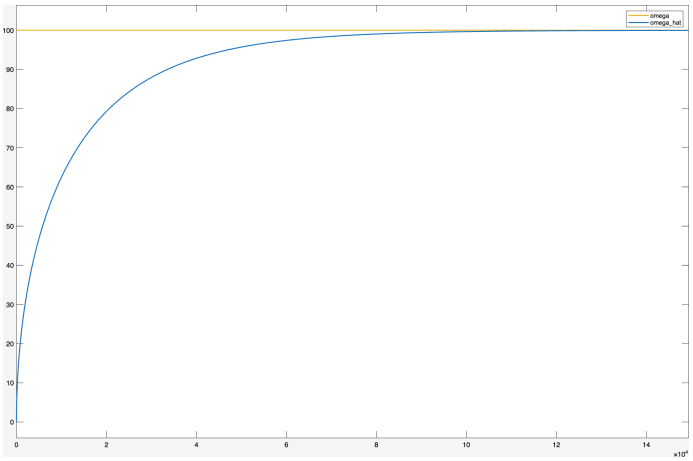
- For $k = 0.1$, it converges in about $1.4 \cdot 10^4 s$ without oscillations.



- For $k = 1$, it converges in about $1.1 \cdot 10^4 s$ without oscillations.



- For $k = 10$, it converges in about $1.3 \cdot 10^4 s$ without oscillations.

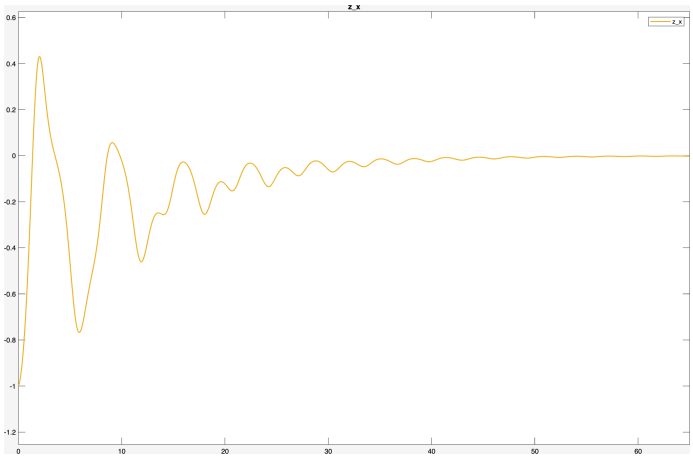


4 $y_2 = \sin t$

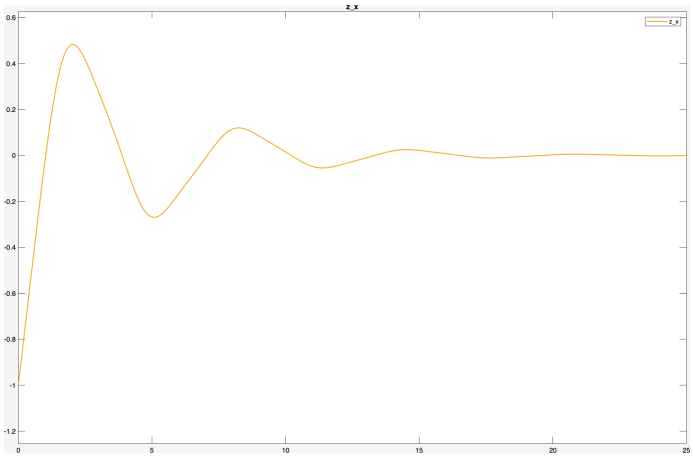
4.1 Estimation error z_x

With a disturbance-free measurement of $y_2 = \sin t$, the estimation error for x, denoted by z_x, behaves as follows.

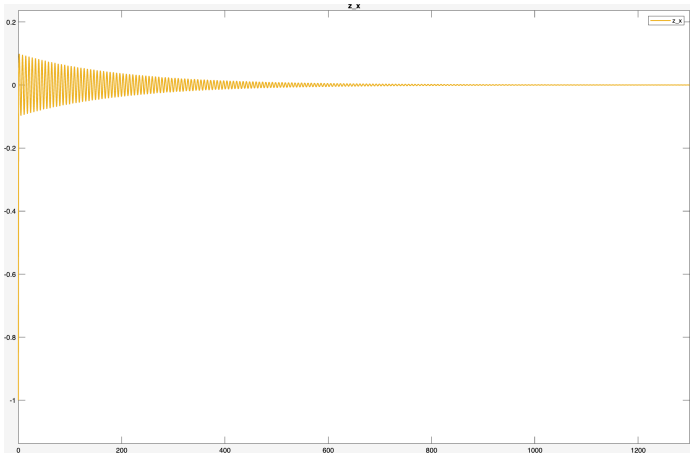
- For $k = 0.1$, it converges in about $60s$, with a maximum overshoot of 0.8.



- For $k = 1$, it converges in about $25s$, with an overshoot of 0.5.



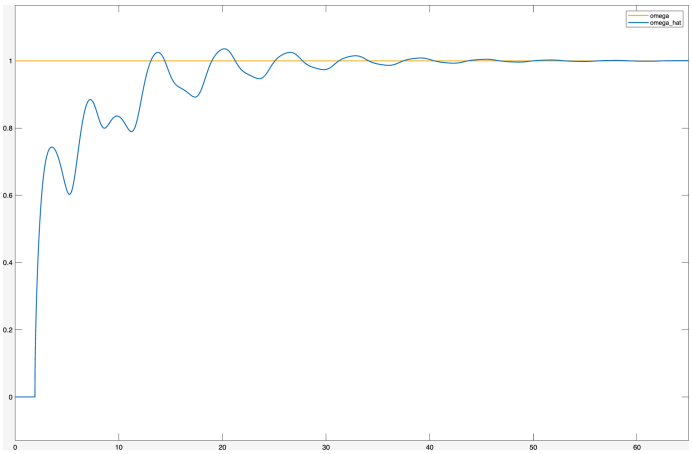
- For $k = 10$, it converges in about $1200s$ after numerous oscillations with a maximum amplitude of 0.1.



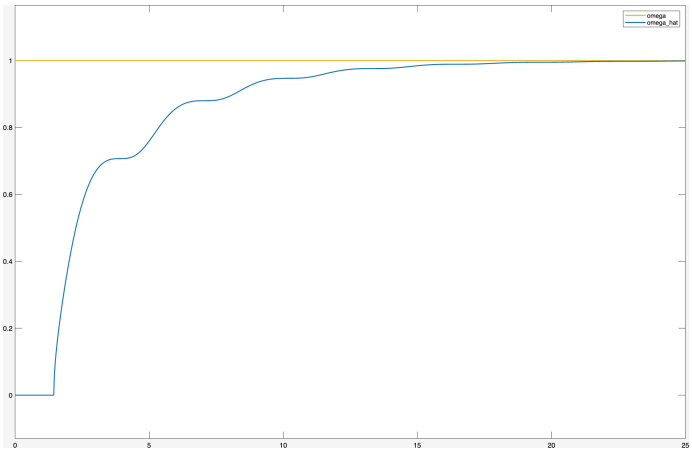
4.2 Estimation of the angular frequency ω

With a disturbance-free measurement of $y_2 = \sin t$, the estimate of the angular frequency ω behaves as follows.

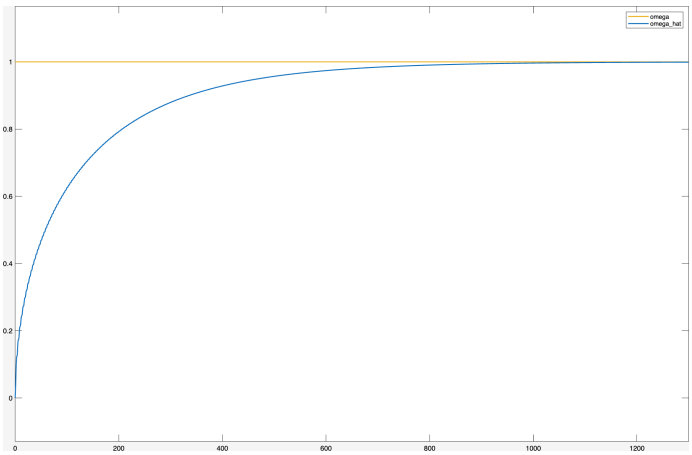
- For $k = 0.1$, it converges in about $60s$, with an overshoot of 1.03.



- For $k = 1$, it converges in about $23s$ without overshoot.



- For $k = 10$, it converges in about $1200s$ without oscillations.

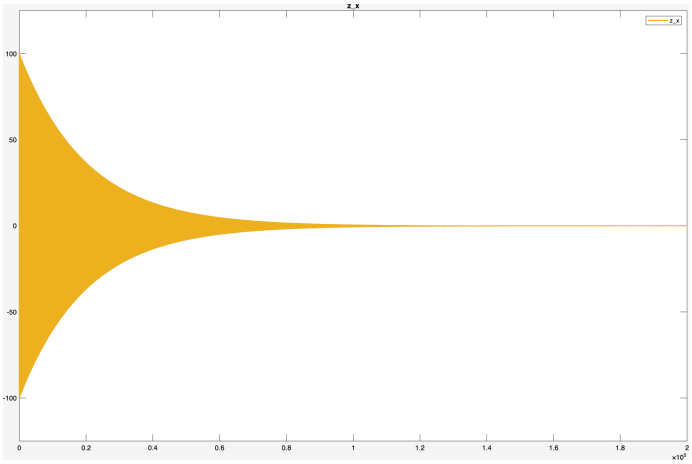


5 $y_m = \sin 100t + 0.001\sin(50t)$

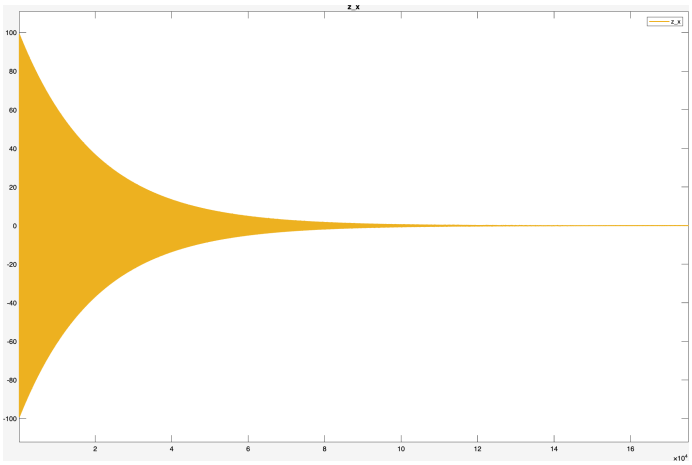
5.1 Estimation error z_x

With a disturbance in the measurement of $y_1 = \sin 100t$, the estimate of the angular frequency ω behaves as follows.

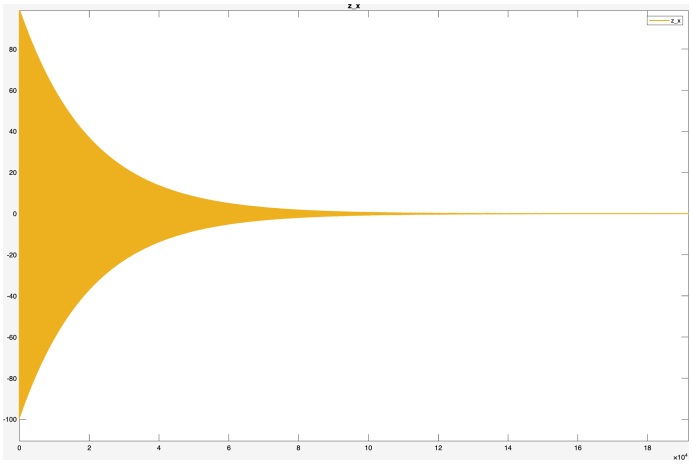
- For $k = 0.1$, it converges in about $1.9 \cdot 10^5 s$, with oscillations reaching an amplitude of 100.



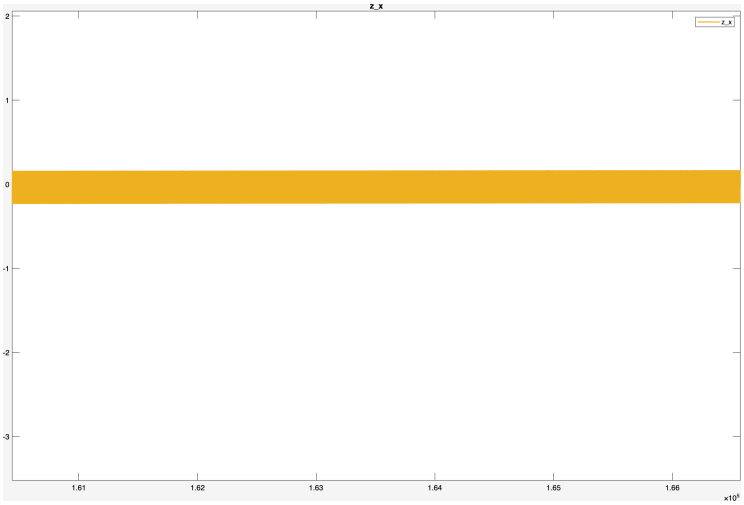
- For $k = 1$, it converges in about $1.5 \cdot 10^5 s$, with oscillations reaching an amplitude of 100.



- For $k = 10$, it converges in about $1.5\,10^5 s$, with oscillations reaching an amplitude of 100.



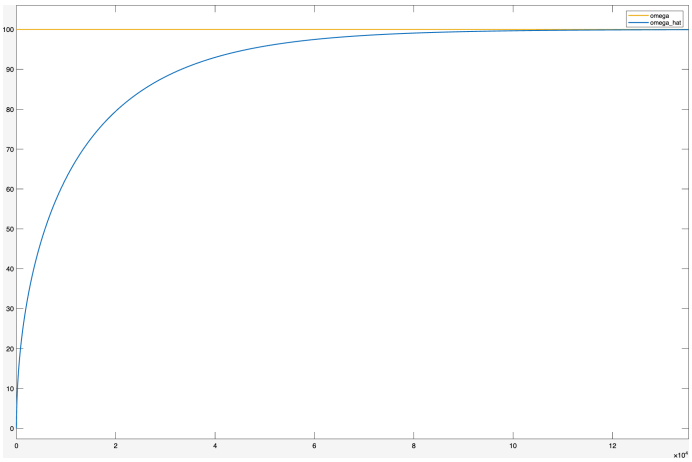
In all three cases, the estimation error z_x continues to oscillate between -0.2 and $+0.2$ in steady state.



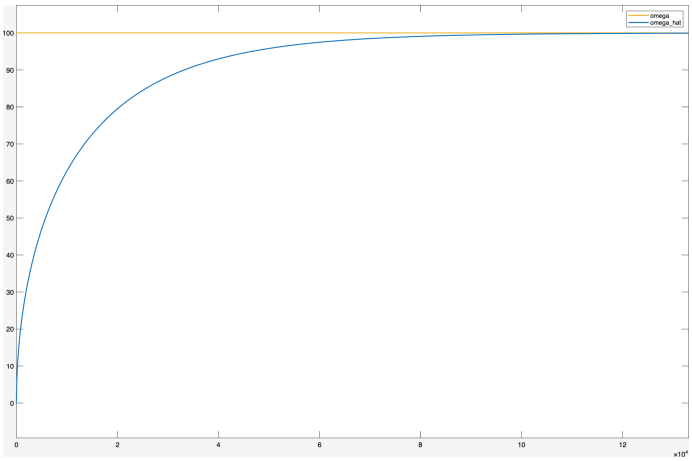
5.2 Estimation of the angular frequency ω

With a disturbance in the measurement of $y_1 = \sin 100t$, the estimation error for x, denoted by z_x, behaves as follows.

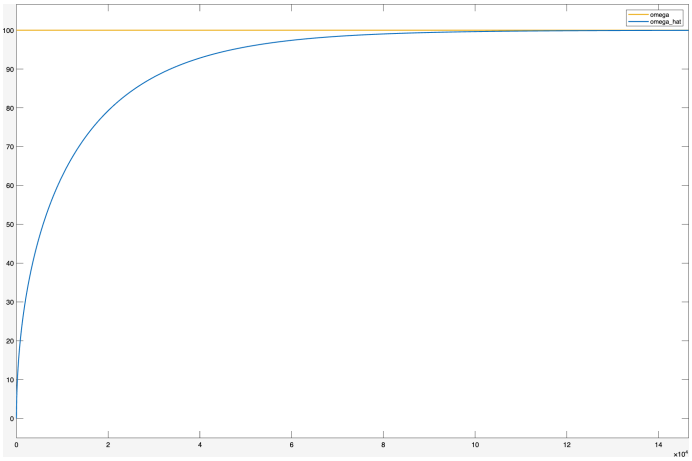
- For $k = 0.1$, it converges in about $1.3s$.



- For $k = 1$, it converges in about $1.2 \cdot 10^5 s$ without oscillations.



- For $k = 10$, it converges in about $1.4 \cdot 10^5 s$ without oscillations.



6 Conclusions

The simulations yielded the following observations.

- The convergence rate depends on both the design parameter k and the angular frequency of the signal being estimated. The best choice of k therefore varies with the signal dynamics.

- For the high-frequency signal $y_1 = \sin(100t)$, the angular frequency estimate takes on the order of 10^5 s to converge for all tested values of k. It exhibits oscillations with a maximum amplitude of approximately 100. The state estimation error z_x converges monotonically, without overshoot.

- For the low-frequency signal $y_2 = \sin t$, a larger k can slow convergence while reducing overshoot in the estimation error. The overshoot disappears for $k = 10$. Of the tested values, $k = 1$ offers the best trade-off between speed and smoothness.

- In the presence of measurement noise, the state estimation error continues to oscillate within approximately ± 0.2. The disturbance propagates through the estimator and prevents the error from converging exactly to zero. The angular frequency estimate is more robust to the disturbance than the state estimate.